



Programming education with ChatGPT: outcomes for beginners and intermediate students

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Abstract

Artificial Intelligence (AI) has rapidly emerged as a powerful tool with the potential to enhance learning environments. However, effective use of new technologies in education requires a good understanding of the technology and good design for its use. Generative AI such as ChatGPT requires particularly well-designed instructions due to its ease of use in asking and answering questions using natural language. This study investigates the effectiveness of ChatGPT in assisting the learning process of university students enrolled in Python programming courses. The primary question addressed is how differently the use of ChatGPT affects two groups at different levels of programming proficiency—beginners and intermediate. The study compares the impact of ChatGPT on academic performance between these two groups. The results show that for beginners, ChatGPT did not significantly contribute to improved learning outcomes. In addition, a survey was conducted to explore students' perceptions of the benefits and challenges of using ChatGPT, which revealed differences in responses between the two groups of learners. This research provides valuable insights into the potential applications of AI tools in educational settings, particularly in the context of programming education.

Keywords Generative AI · Education Tool · ChatGPT · Programming Education · Prior Knowledge Activation · Proficiency level

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1 Introduction

The landscape of education has recently undergone a rapid transformation, driven by technological advancements. Foremost among these innovations is artificial intelligence (AI), which has had a significant influence (Kowitlawakul et al., 2017; Logan et al., 2021). AI's integration into education has created opportunities for improving learning environments including enhanced learning comprehension, flexible scheduling and control over students' learning paths, along with accelerated development of users (Chang et al., 2023; Kabudi et al., 2021; Moreno-Guerrero et al., 2020).

In recent years, Generative AI, which encompasses artificial intelligence models that can generate new content, has seen wide-ranging applications. The term “generative” indicates that these systems do more than just analyze or process data; they create new data that previously did not exist (Baidoo-Anu & Owusu Ansah, 2023). These applications include creating realistic images and video, composing music, creating textual content, and writing computer code (Chen et al., 2023; Zinovieva et al., 2021). Among the applications of Generative AI, ChatGPT has gained significant popularity (Budhwar et al., 2023; Kalla & Smith, 2023). This language model has the capability of generating code from natural language descriptions, allowing users to create simple programs even without previous programming experience (Adamson & Bägerfeldt, 2023; McManus, 2023).

Programming education has a multitude of positive effects that extend beyond the realm of computer science and technology (Russell & Norvig, 2016). It cultivates a skill set that is increasingly important in a variety of professional fields and daily life, fostering analytical thinking, creativity, and a better understanding of the technology-driven world, such as problem-solving skill, logical thinking, persistence and resilience, digital literacy, and empowerment (Agbo et al., 2019; Gordon et al., 2022; Siegfried et al., 2021; Tian et al., 2023; Yilmaz & Yilmaz, 2023a). The connection of programming's importance and Generative AI has spawned the emerging field of automatic code generation, which was further accelerated by the introduction of ChatGPT by OpenAI on November 30, 2023 (OpenAI, 2023). The advantage of ChatGPT is that it provides immediate, well-organized answers to user questions. However, this could make it difficult to develop the positive effects often achieved through programming, such as analytical thinking, problem solving, logical reasoning, and the development of perseverance and resilience. Therefore, the emerging question is: “Is the use of ChatGPT helpful in learning programming?”

Prior knowledge activation is a teaching strategy that involves activating prior knowledge before introducing new concepts. It is based on the understanding that connecting new information to existing knowledge structures enhances learning (Pardo, 2004). The assistance provided by ChatGPT may be limited for beginners, because they lack prior programming knowledge. This makes it difficult for them to connect new information with existing concepts. In contrast, intermediate learners who already have a foundational understanding of programming can

more easily integrate new insights from ChatGPT with their existing skills, leading to more efficient learning. Therefore, ChatGPT may not be helpful for beginners who have little prior knowledge, but it may be helpful for intermediate students who have more prior knowledge.

Studies have been conducted on how ChatGPT has served as an educational tool for students and educators, including recent research in the field of programming (Kasneji et al., 2023; Lo, 2023; Qureshi, 2023; Yilmaz & Yilmaz, 2023b). However, there is a lack of research that examines the effect of ChatGPT on students' learning in actual classes by their learning level. Thus, the purpose of this research is to investigate the impact of using ChatGPT on programming learning to different learning level of student group, and how the use of ChatGPT can improve programming learning.

The study aims to answer two main questions:

- (1). Does ChatGPT serve as a useful tool for learning programming for both beginners and intermediates?
- (2). What are the advantages and disadvantages of studying with ChatGPT for students?

2 Literature review

2.1 Prior knowledge activation

Prior knowledge activation refers to an educational approach where learners are encouraged to recall or utilize their existing knowledge about a topic before being introduced to new content (Haleem et al., 2022; Henderson, 2007; Schmidt et al., 2011). This strategy is grounded in that learning is more effective when new information is connected to what is already known. It involves asking questions, prompting discussions, or engaging in activities that bring learners' attention to what they have previously learned that is relevant to the new material (Boekaerts, 1997; Chang et al., 2023; Singh et al., 2023).

The motivation behind prior knowledge activation is following. First, it helps to enhance comprehension by creating links between old and new knowledge, making the new content more relatable and easier to understand. Second, it provides a context for the new information, which can improve retention and recall. Third, it fosters a more active learning environment by encouraging students to contribute their knowledge and experiences. Fourth, it identifies misconceptions that may interfere with learning new information (Bellana et al., 2021; Cordova et al., 2014; Rivet & Krajcik, 2008; Singh et al., 2023; Tajeddin & Soudabeh, 2016).

It is not a standalone strategy or theory but rather a principle or instructional strategy that stems from various theories of cognitive psychology and educational psychology. The concept is deeply rooted in several established theories (Deubel, 2003; Young et al., 2014). First one is Schema Theory. According to schema theory, learners interpret information in the context of their existing cognitive structures (schemas). Activating prior knowledge helps learners to connect new information

to these existing schemas, facilitating better comprehension and retention (Anderson et al., 1977). Second one is Constructivist Theory. Constructivist learning theory, particularly as developed by Jean Piaget, suggests that knowledge is constructed by building upon what is already known. Activating prior knowledge is a way to prepare the mind for adding new knowledge to these constructions (Kumar, 2006). Last one is Cognitive Load Theory. This theory posits that learning is more effective when unnecessary cognitive load is reduced. By activating what learners already know, instructors can help reduce the intrinsic cognitive load associated with processing new and possibly complex information (Kirschner, 2002).

To examine whether prior knowledge activation applies to ChatGPT and programming education, let's consider beginners new to programming. The assistance from ChatGPT might be limited for them because they lack any prior knowledge about programming, making it challenging to connect past knowledge with new information. However, for intermediate-level learners who already possess foundational programming knowledge, the connection of their existing skills with new insights from ChatGPT can lead to more efficient learning. Ultimately, this blend of past knowledge and new information from ChatGPT proves beneficial for their educational progress.

2.2 ChatGPT

ChatGPT distinguishes itself from conventional programming environments and tools through its AI language model, which facilitates interaction with users in natural language (Chang et al., 2023; Yilmaz & Yilmaz, 2023a, b). As a result, even individuals lacking programming expertise can easily solve programming problem using ChatGPT. (Surameery & Shakor, 2023). There are other generative AIs that are more suitable for programming, such as GitHub Copilot.¹ However, GitHub Copilot is basically a paid service with some exceptions, and ChatGPT is much more popular with over 180.5 million users by the end of 2023,² so ChatGPT was used in the study.

According to Ray (2023), ChatGPT offers several distinct advantages over traditional programming tools, thanks to its capabilities as an AI language model. One key feature is its ability to understand natural language. This makes it accessible even to those without formal programming training, enabling them to solve programming problems. ChatGPT acts almost like a real-time tutor. It can help write and explain code, demonstrate how to write code more accurately, and provide debugging assistance, which builds confidence in programming. Furthermore, ChatGPT's problem-solving assistance in various domains, including programming, helps users develop analytical and critical thinking skills. Finally, ChatGPT aids students in advancing to more sophisticated stages in programming, enabling them to reach higher levels of proficiency. It also offers solutions for interdisciplinary issues

¹ <https://github.com/features/copilot>

² <https://www.namepepper.com/chatgpt-users>

like math and physics, which are integral to the programming process (Choudhury & Shamszare, 2023; Roumeliotis & Tselikas, 2023; Singh et al., 2023; Yilmaz & Yilmaz, 2023b).

However, ChatGPT has limitations that can hinder its effectiveness. For example, it has a tendency to make assumptions about what the user means rather than asking for more details, to be influenced by how questions are worded and repeated, and to provide answers that seem credible but are actually incorrect or meaningless. These erroneous responses are commonly termed “hallucinations,” stemming from the absence of a reliable source of truth (Adamson & Bägerfeldt, 2023; OpenAI, 2023). From a more macroscopic perspective, ChatGPT could have disadvantages for its users. According to the study by Yilmaz & Yilmaz (2023b), using ChatGPT for learning programming may present the following limitations for students: ChatGPT could make students lazy, potentially negatively impact their analytical thinking skills, and reduce their potential. It might not always provide accurate answers, leading to responses that fall short of expectations. It could make students passive, especially if ChatGPT’s code is so advanced that they cannot understand it on their own. Furthermore, reliance on ChatGPT might lead to negative decisions in future career choices for students. This research suggests that these limitations or disadvantages of ChatGPT may apply differently to beginners and intermediates in programming.

3 Research method

3.1 Procedures

This research is based on an experimental cross-sectional design. This study was conducted at a four-year university in South Korea during the fall semester of 2023, from September 4 to December 16. Two courses were chosen to get a data from beginners and intermediates: Introductory Programming and Practice; and Understanding Machine Learning and Deep Learning. Both courses use Python as the programming language. Python has a concise and easy-to-understand syntax and is widely used in a variety of fields, including data science, artificial intelligence, and web development. (Ayer et al., 2014; Pellet et al., 2019).

Introductory Programming and Practice is designed for first-year students with no programming experience. The left column of Table 1 shows the topics covered in the beginners’ course. Among 120 students enrolled in an introductory programming principles and practice course, 80 voluntarily chose to participate in this study. These beginners had never taken a programming course before. In the first class, the professor introduced the study and conducted a preliminary survey among the students who were interested in participating. According to this survey, while all the students had experience using ChatGPT, none had used it for programming. However, they were aware that ChatGPT could be used to copy homework assignments or to identify bugs in the code they input. The 80 participants were individually randomized into two groups, where 40 students were allowed to use ChatGPT and the other group was not allowed to use ChatGPT.

Table 1 Weekly lecture topics of two courses

Week	Beginners' Course	Intermediate Course
1	Chapter 1: Python Introduction Chapter 2: Python Installation and Programming Environment	Course Introduction Useful Python Features
2	Chapters 3 and 4: Python Grammar—Identifiers and Reserved Words	Web Data Structures and Web Scraping/Crawling I
3	Continuation of Chapters 3 and 4 Chapter 5: Basic Data Types	Web Data Structures and Web Scraping/Crawling II
4	Continuation of Chapter 5 Chapter 6: Compound Data Types	Basics of Text Analysis with NLTK
5	Continuation of Chapter 6	Basics of Text Analysis with NLTK
6	Further Study of Chapter 6	Basics of Korean Text Analysis
7	Completion of Chapter 6 Mid-term Examination Q & A Session	Basics of Machine Learning and Deep Learning
8	Mid-term Examination (<i>Supervised</i>)	Mid-term Examination (<i>Supervised</i>)
9	Chapter 7: Iterating Operations	Image Processing Based on Deep Learning Models
10	Chapter 8: Control Flow Statements	Image Processing Based on Deep Learning Models
11	Continuation of Chapter 8	Text Processing Based on Deep Learning Models
12	Chapter 10: Functions	Text Processing Based on Deep Learning Models
13	Continuation of Chapter 10	Time Series Data Analysis Based on Deep Learning Models
14	Course Wrap-up Final Examination Q&A Session	Time Series Data Analysis Based on Deep Learning Models
15	Final Examination (<i>Supervised</i>)	Final Examination (<i>Supervised</i>)

Understanding Machine Learning and Deep Learning is designed for graduate students in Electrical Information or Computer Engineering departments, who have completed at least three programming courses and consider themselves proficient in programming. Participation was voluntary, and out of over 60 enrolled students, thirty randomly selected students were allowed to use ChatGPT and other students were not. The right column of Table 1 shows the topics covered in the intermediate course.

Surveys were conducted after the third assignment, only with the students that were allowed to use ChatGPT. Few studies have examined students' perspectives on the use of ChatGPT for learning programming (Yilmaz & Yilmaz, 2023a). Understanding students' viewpoints on using ChatGPT in programming education can offer educators and curriculum developers valuable insights, particularly regarding its integration into the course. The purpose of the survey is to explore whether the benefits and potential disadvantages or concerns of using ChatGPT differ between beginners and intermediates. The survey consists of four sections in total. The first section includes demographic information. The second section lists the advantages of ChatGPT as described by OpenAI, as well as the benefits specifically related to programming with ChatGPT. The third section contains advantages of using ChatGPT, and the fourth section covers disadvantages of using ChatGPT. The detailed contents available in the appendix.

3.2 Measurements

To measure students' learning outcomes, two sets of materials were developed for research. This study presents them below in four subsections: assignment, and examination.

3.2.1 Assignment

A total of four homework assignments were given to both beginners and intermediates. The first assignment for each group is shown in Figs. 1 and 2. The scores of assignments can serve as an indicator of improved programming skills. However, since students can receive various forms of assistance for assignment, they may not be the most accurate measure. Therefore, measurements under controlled conditions are necessary for a more accurate assessment.

Assignment 1. Pythagorean theorem

Write a program that takes the lengths of two sides of a right triangle except for the hypotenuse and finds the area of the right triangle and the length of the hypotenuse.

- Use the function `input()` to get an integer as input .
- Print the result using either `str.format()` or f-string

The implementation of the program should follow the following rules.

1. Using the prompt as follows, get length A of the first side with a right angle and then length B of the second side with a right angle. A and B are natural numbers.

Enter the length of the first side with a right angle : A
 Enter the length of the second side with a right angle : B
2. Result will be following format.

Area is X, the length of hypotenuse is Y

Note. To calculate the square root, we can use the operator `**`.

e.g. `4 ** 0.5 = 4`

Example.

Enter the length of the first side with a right angle : 3
 Enter the length of the second side with a right angle : 4
 Area is 6.0, the length of hypotenuse is 5.0

Fig. 1 First assignment: beginner level class

Assignment 1. Wikipedia

Through the lessons in weeks 2 and 3, we have learned various ways to explore information on web pages using Python. In particular, during the third week, we extracted the contents of a specific Wikipedia page and saved it as a text file. Building on this, select a topic (keyword) you are interested in and write a code to save the content of the Wikipedia page about this keyword as a text file.

Assignment Deliverables

- A text file with the extracted content of the Wikipedia page.
- A Python code file (.py) that extracts and saves the content from the Wikipedia page.

Note.

The text file should be submitted as 'wiki-[keyword].txt', and the Python code file should be submitted as '[student_id].py.' Students must submit a Python file that can be executed directly with the command 'python [student_id].py', not a Jupyter notebook file.

Fig. 2 First assignment: intermediate level class

3.2.2 Examination

In-class exams conducted in controlled settings are more accurate indicators of students' programming improvement. It is forbidden to use ChatGPT during the exams. Before the exam, a monitoring program was installed to record students' keystrokes. After the exam, the recordings were reviewed to check for any suspicious recordings such as 'Ctrl + C(copy)', 'Ctrl + V(paste)' or prompt commands. If students regularly reviewed lessons on their own and earnestly attempted to solve problems of assignments, they would have found the exams not too challenging. For both beginner and intermediate levels, the exams were designed by slightly modifying or increasing the difficulty of the lab exercises during the class and assignments.

The content of the midterm exam is listed in the Figs. 3 and 4. If the group using ChatGPT relied on it to do their assignments, either by asking it questions in natural language or by writing code hastily and then using ChatGPT to identify and fix bugs before submission, they would have found the exams extremely challenging. Conversely, if they exerted effort in coding themselves, consulted ChatGPT afterwards, compared its suggestions with their own code, and submitted improved versions, the exams would not have been as difficult for them.

Mid-term Exam	Problem 2. Operation profit calculation (8 points)
Problem 1. Word extraction (7 points) Extract words from a movie review using string methods. 1) Assign the following review to a single string variable. Review is graded with a number between 0 and 9, and hyphen (-). <pre>8 - Denis Villeneuve film. It was such an incredible technical achievement that I was blown away by it over and over again. I admire any director who is able to create a self-contained world on such a scale, but this is an epic for the eye, not the heart.</pre> 2) Use <code>str.strip()</code> to remove the rating from the review 3) Use <code>str.replace()</code> to remove the symbol characters(' ', '.', and '-') and the following stop words (words that do not have much meaning). 'It', 'was', 'the' 4) Use <code>str.split()</code> to split the review based on the white space. 5) Use <code>str.join()</code> to combine the separated words into a CSV (Comma Separated Values) string connected with commas (,) and print them. Example. <pre>> python midterm_1.py Denis,Villeneuve,film,such,an,incredible,technical,achievement,that,I, blown,away,by,it,over,and,over,again,I,admire,any,director,who,is,able ,to,create,a,selfcontained,world,on,such,a,scale,but,this,is,an,epic,f or,eye,not,heart</pre>	Write a program to calculate gross profit and operating profit using data from given financial statements. Data are given in the form of quarterly (Revenue, COGS-Cost of Goods Sold, Operating expenses). The sequence of the program is as follows. 1) Allocate the following 4 tuple variables: Q1, Q2, Q3, Q4. These store the values of quarterly revenue, COGS, Operating expenses respectively. <pre>Q1 = (7760, 6290, 450) Q2 = (8920, 7800, 500) Q3 = (8145, 7600, 485) Q4 = (7105, 5920, 450)</pre> 2) Reconstruct the given data into dictionary as follows and print it. <pre>{'Revenue': (7760, 8920, 8145, 7105), 'COGS ': (6290, 7800, 7600, 5920), 'Operating expenses': (450, 500, 485, 450)}</pre> 3) Find the quarterly gross profit and operating profit, add them to the dictionary, and print the updated dictionary. Note. Gross profit = Revenue - COGS Operating profit = Gross profit - Operating expenses. Example. <pre>> python midterm_2.py {'Revenue': (7760, 8920, 8145, 7105), 'COGS ': (6290, 7800, 7600, 5920), 'Operating expenses': (450, 500, 485, 450)} {'Revenue': (7760, 8920, 8145, 7105), 'COGS ': (6290, 7800, 7600, 5920), 'Operating expenses': (450, 500, 485, 450), 'Gross profit': (1470, 1120, 545, 1185), 'Operating profit': (1020, 620, 60, 735)}</pre>

Fig. 3 Midterm examination: beginner level class

Mid-term Exam
Practical Coding Tasks (20 points) 1. Advanced Python Scripting: - Write a Python script that uses list comprehensions to filter and process data from a given list. - Create a context manager in Python that handles the opening and closing of a file. 2. Web Scraping/Crawling Project: - Develop a Python script to scrape data from a given website (e.g., extracting all headings from a blog page). The script should handle exceptions and export the data to a CSV file. - Write a Python script using Selenium to automate and log in to a website, then perform a specific task (like searching for a term or navigating through pages). Bonus Question (Extra Credit): - Illustrate how you would approach scraping a website with dynamically loaded content using AJAX. Discuss the tools and libraries you would use and the challenges you might face. Exam Guidelines: - The practical section requires coding on a computer, with a time limit of 90 minutes. - For coding tasks, students must comment on their code to explain their logic and approach. - All tasks and questions must be completed individually.

Fig. 4 Midterm examination: intermediate level class

3.2.3 Expert evaluation

Three professors from the departments of Business Administration and Computer Science carefully reviewed assignments and exams of two courses based on the criteria in Table 2. The purpose of this evaluation was to explore whether the use of ChatGPT could lead to unique aspects in students' work. ChatGPT tends to provide optimal answers based on the user's natural language queries, utilizing its database rather than considering the user's background knowledge. Therefore, it might introduce ideas or approaches that go beyond what was taught in the class. Based on the Prior Knowledge Activation theory, beginner, due to their lack of or absence of existing knowledge, may find it challenging to create a synergy effect through the activation between new information and their existing knowledge. When beginners use ChatGPT to get answers to their questions, they might not realize that these answers surpass what they have learned in class.

4 Results

4.1 Evaluation of learning outcomes

An analysis was conducted to confirm if the use of ChatGPT is helpful in learning programming and if the effectiveness varies depending on the prior knowledge level of students. Learning outcomes were measured in two ways: assignments and exams. The scores of each group, based on their prior knowledge level and whether they used ChatGPT or not, are shown in Table 3. The scores for the first and second assignments were combined, and the scores for the third and fourth assignments were also combined. Each was originally worth 15 points for the Beginner class midterm exam, 20 points for the Intermediate class midterm exam, 25 points for the Beginner class final exam, and 40 points for the Intermediate class final exam, but all points, including assignment points, were converted to a 10-point scale.

The t-test results showed a significant difference in performance on the midterm exam based on the use of ChatGPT in a group of beginner level students ($t = -3.333$, $p < 0.01$), and final exam as well ($t = -2.030$, $p < 0.05$). This difference was not observed for the assignments or exams for intermediate students. Considering that students can solve assignments with the help of ChatGPT or the textbook, but not exams, it can be concluded that the use of ChatGPT for beginner students is not conducive to their learning.

Average scores of each learning outcomes with 95% confidence interval are shown in Fig. 5. For beginner students, there is a clear difference between using ChatGPT on the exams, but there is no significant difference in the assignments. Interestingly, the difference in performance on the midterm is not observed on the post-midterm assignments, but reappears on the final exam. This observation suggests that students using ChatGPT do not improve their programming skills well, but they do become more proficient at problem solving with ChatGPT.

Table 2 Expert evaluation criteria

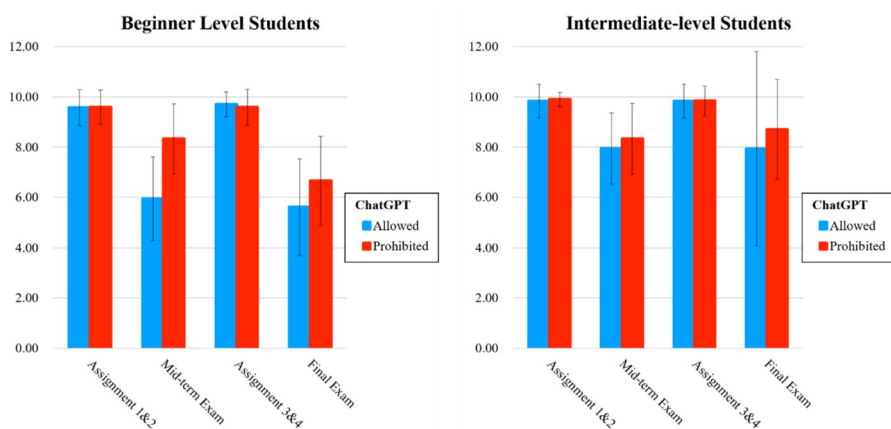
Criteria	Definitions	Reference
Accuracy	how precisely and correctly the code meets its intended purpose and specifications	Adamson & Bägerfeldt, 2023; Dromey, 1995
Readability	how easy it is for a person to read and understand the source code of a program	Adamson & Bägerfeldt, 2023; Dromey, 1995
Quality	how well the source code adheres to good programming practices, making it efficient, error-free, maintainable, and easily understandable	Adamson & Bägerfeldt, 2023; Dromey, 1995
Creativity	how can one demonstrate the ability to approach problems and tasks in innovative and original ways within the realm of software development	Komm et al., 2020
Reflection (Only Assignment)	how to successfully incorporate what was learned that week into the assignment	Chang, 2019

Table 3 Results of analysis examining the effect of ChatGPT on learning outcomes

Logistic parameter	ChatGPT allowed		ChatGPT prohibited		t	p	Cohen's d
	M	SD	M	SD			
Beginner Level							
Assignment 1 & 2	9.575	1.124	9.588	1.074	− 0.050	0.735	0.011
Mid-term Exam	5.950	3.457	8.333	2.916	− 3.333	0.001**	0.745
Assignment 3 & 4	9.700	0.775	9.588	1.018	0.690	0.492	0.154
Final Exam	5.610	2.399	6.660	2.220	− 2.030	0.046*	0.454
Intermediate Level							
Assignment 1 & 2	9.833	0.913	9.900	0.381	− 0.368	0.715	0.095
Mid-term Exam	7.950	1.891	8.333	1.895	− 0.785	0.436	0.202
Assignment 3 & 4	9.833	0.913	9.850	0.458	0.378	0.707	0.097
Final Exam	7.942	2.583	8.708	1.333	− 1.440	0.078	0.373

Mean parameter values for each of the analyses are shown for ChatGPT allowed beginner students ($n = 40$), ChatGPT prohibited beginner students ($n = 40$), ChatGPT allowed intermediate-level students ($n = 30$), and ChatGPT prohibited intermediate-level students ($n = 30$), as well as the results of *t* tests comparing the parameter estimates between the use of ChatGPT

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

**Fig. 5** Average scores of each learning outcomes with 95% confidence interval

4.2 Expert evaluation

The experts examined the difference in the quality of the materials submitted by the students. For beginners' assignments, since the content was not very difficult, aspects such as accuracy, readability, and quality were similar regardless of the use of ChatGPT. However, two peculiar points were noted in their homework. Firstly, the group using ChatGPT submitted work that included content beyond what was

```

A = int(input('Enter the length of the first side with a right angle: '))
B = int(input('Enter the length of the second side with a right angle: '))

C = A ** 2 + B ** 2
area = 0.5 * A * B

print('Area is {}, the length of hypotenuse is {}'.format(area, C))

```

Fig. 6 Typical answer for first homework: beginner level class

```

import math

a = int(input("Enter the length of the first side with a right angle: "))
b = int(input("Enter the length of the second side with a right angle: "))

c = math.sqrt(a**2 + b**2)

area = 0.5 * a * b

print("Area is {}".format(c))
print("the length of hypotenuse is {}".format(area))

```

Fig. 7 Answer using math module for first homework: beginner level class

taught in class. For instance, the first assignment was merely about using basic data types, yet a student, who had not learned about the math module, used it in their submission. Unlike the submission in Fig. 6, which was solved using operators (e.g., `**`) taught in class and mentioned in the assignment description, the submission in Fig. 7 included packages (e.g., `math`) and methods (e.g., `sqrt`) that were not covered in class. A similar phenomenon is observed in the submission for Fig. 8. This submission includes a user-defined function that has not yet been taught, and it employs the technique of using variable names directly in an f-string. This approach is significantly different from the submission in Fig. 9, which strictly adhered to what was taught in class.

The second notable aspect was the name of variables. Beginners might not be aware of the implicit conventions used by developers in naming variables. However, in the assignment, experts noticed some students using variable names that seemed like those typically used by developers. For instance, in the homework assignment where students were asked to write a while loop to find the least common multiple (LCM) of three integers, experts observed that students who used ChatGPT set up their variables in a way like `lcm_num_1`. This difference highlights the influence of ChatGPT in guiding students towards more appropriate and meaningful variable naming, aligning with the task at hand. It suggests that ChatGPT, by providing context-aware suggestions, may enhance the coding practices of beginners who are still learning to navigate the nuances of programming languages like Python.

In the case of exams for beginners, experts did not observe any difference in performance based on the criteria above, regarding the use of ChatGPT. The intended learning outcome (ILO) of the beginner course is to acquire a knowledge base for

```

def calculate_grade(midterm_score, final_score):
    if midterm_score >= 60:
        if final_score >= 80:
            return 'A+'
        elif final_score >= 70:
            return 'A0'
        elif final_score >= 60:
            return 'A-'
        else:
            return 'C+'
    else:
        if final_score >= 90:
            return 'A-'
        elif final_score >= 80:
            return 'B+'
        elif final_score >= 70:
            return 'B0'
        else:
            return 'C+'

midterm_score = int(input("What is your midterm exam score? "))
final_score = int(input("What is your final exam score? "))

grade = calculate_grade(midterm_score, final_score)
print(f"Your final grade is {grade}")

```

Fig. 8 Answer using function module for third homework: beginner level class

```

midterm = int(input('What is your midterm exam score? '))
final = int(input('What is your final exam score? '))

result = 'Your final grade is {}'

if midterm >= 60:
    if final >= 80:
        print(result.format('A+'))
    elif 80 > final >= 70:
        print(result.format('A0'))
    elif 70 > final >= 60:
        print(result.format('A-'))
    else:
        print(result.format('C+'))

if midterm < 60:
    if final >= 90:
        print(result.format('A-'))
    elif 90 > final >= 80:
        print(result.format('B+'))
    elif 80 > final >= 70:
        print(result.format('B0'))
    else:
        print(result.format('C+'))

```

Fig. 9 Typical answer for third homework: beginner level class

writing Python program code. Students using ChatGPT seem to achieve this ILO based on the clean, neat, and optimized code they produce in their assignments. However, supervised in-class exam results show that it is not.

In the evaluation of assignments for intermediate-level students, the code submitted by the group using ChatGPT was particularly impressive. Frankly, it was evaluated as being flawless. The example is shown in Fig. 10. The submission in Fig. 10 defines classes and functions to solve the given task. However, there is some over-implementation of features that go beyond the requirements. For instance, it includes defining methods inside the class that are not necessary and requesting to type in a keyword. The objective of assignment 3 is to write code that scrapes a Wikipedia page for a given keyword.

In the case of intermediate-level exams, experts observed some differences based on the usage of ChatGPT. Exams from the group that used ChatGPT showed slightly higher readability and creativity in their code. This is thought to be because intermediate students are already well-versed in Python. They can interpret code and distinguish between high-quality, low-quality, unique, and unoriginal codes. Regular use of ChatGPT might have exposed them to high-quality code, which was then reflected in their exams. The ILO of intermediate course is to use and apply external libraries to Python program code. Students' assignments using ChatGPT have some elements of over-implementation, although the consistent scores on supervised in-class exams show that they are meeting the ILO.

4.3 Responses of participants

Respondents were asked which aspect of ChatGPT they found most appealing. Multiple responses were allowed, and eight options were provided: Cost, Time Efficiency, Infinite Creativity, Consistency, Ease of Use, High-quality, Flexibility, and Accuracy. The differences in responses between the student groups are shown in Fig. 11. From Fig. 11, beginner level students found ChatGPT most attractive in terms of cost, ease of use, and time savings. Intermediate level students, on the other hand, chose accuracy as the most attractive aspect of ChatGPT. In other words, intermediate level students find it easy to understand how accurate ChatGPT's

```

from html.parser import HTMLParser
import urllib.request

class OnlyBodyContentTextParser(HTMLParser):
    def __init__(self):
        super().__init__()
        self.data = []
        self.in_body_content = False
        self.in_paragraph = False

    def handle_starttag(self, tag, attrs):
        if tag == 'div' and dict(attrs).get('id') == 'bodyContent':
            self.in_body_content = True
        if self.in_body_content and tag == 'p':
            self.in_paragraph = True

    def handle_endtag(self, tag):
        if self.in_body_content and tag == 'p':
            self.in_paragraph = False

    def handle_data(self, data):
        if self.in_paragraph:
            self.data.append(data.strip())

def get_wikipedia_html_content(keyword):
    url = f"https://ko.wikipedia.org/wiki/{urllib.parse.quote(keyword)}"
    with urllib.request.urlopen(url) as response:
        html = response.read().decode('utf-8')
    return html

def main():
    keyword = input("Enter a keyword to search for in Wikipedia : ")

    wikipedia_html = get_wikipedia_html_content(keyword)

    parser = OnlyBodyContentTextParser()
    parser.feed(wikipedia_html)
    content = ''.join([piece for piece in parser.data if piece])

    filename = f"wiki-{keyword}.txt"
    with open(filename, "w", encoding="utf-8") as file:
        file.write(content)

    print(f"The content has been saved to the file {filename}.")

if __name__ == "__main__":
    main()

```

Fig. 10 First homework: intermediate level class

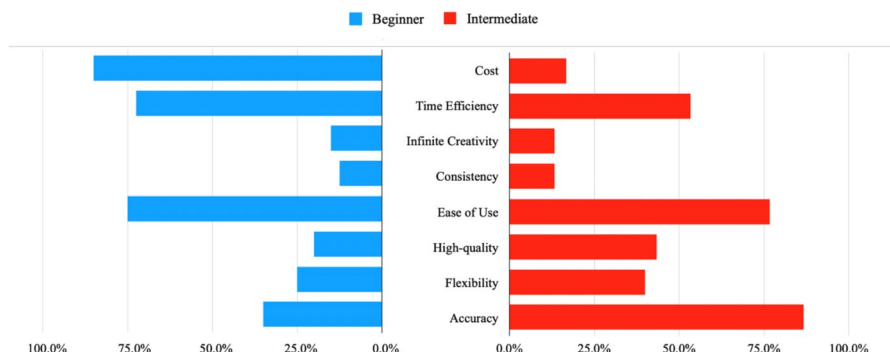


Fig. 11 Frequency distribution of answers by students group

answers are, so they are attracted to the accuracy of ChatGPT. Beginner level students are attracted to ChatGPT because it's very easy to get solutions that look like correct answers.

Along similar lines, the most compelling feature of ChatGPT in the context of programming was asked. Multiple responses were possible, with 10 options available. The differences in responses between the student groups are shown in Fig. 12. Beginner level students chose quick response, easy access, and ease of use (i.e., using natural language) as their top choices. In contrast, intermediate level students preferred examples and applications, and debugging and feedback. Despite differing perceptions of how ChatGPT can be used and its benefits based on prior knowledge, both groups of students recognize the advantages of using ChatGPT.

Twelve questions about the positive aspects of using ChatGPT were asked. The detailed questions are described in Appendix 1. Respondents answered each question on a 5-point scale ranging from strongly agree (1) to strongly disagree (5). The means of the responses by student group are shown in Fig. 13, and the t-test results

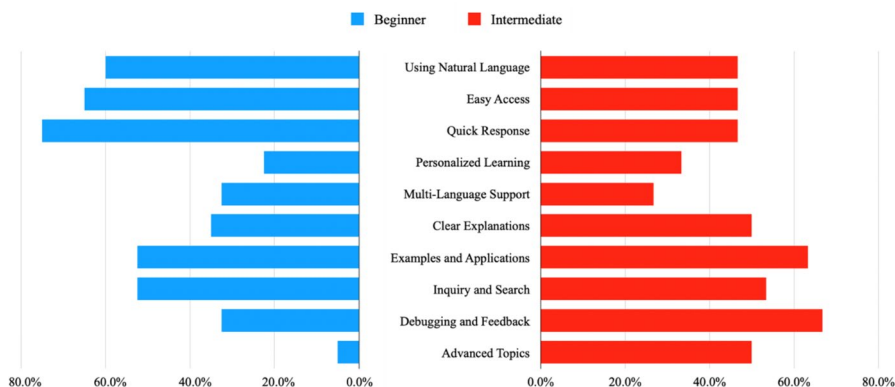


Fig. 12 Frequency distribution of answers by students group

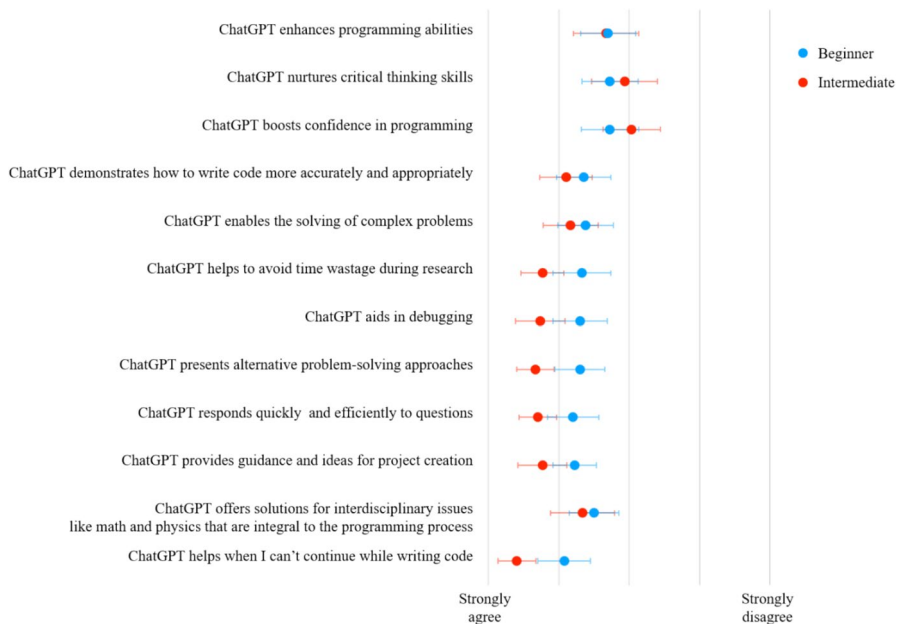


Fig. 13 Perception of advantages of ChatGPT by student group

Table 4 Results of student group comparison on positive perception of ChatGPT

Questions	Beginner Level		Intermediate Level		Variance Assump-tion	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
A01	2.700	1.224	2.667	1.241	Equal	0.911
A02	2.725	1.240	2.933	1.258	Equal	0.492
A03	2.725	1.281	3.033	1.098	Equal	0.294
A04	2.350	1.210	2.100	0.995	Equal	0.360
A05	2.375	1.234	2.167	1.053	Equal	0.460
A06	2.325	1.289	1.767	0.817	Unequal	0.031*
A07	2.300	1.203	1.733	0.944	Equal	0.037*
A08	2.300	1.114	1.667	0.711	Unequal	0.005**
A09	2.200	1.137	1.700	0.702	Unequal	0.027*
A10	2.225	0.974	1.767	0.935	Equal	0.052
A11	2.500	1.109	2.333	1.213	Equal	0.552
A12	2.075	1.163	1.400	0.724	Unequal	0.004**

The assumption of equal variances is confirmed using the F-test with a significance level of 0.05. Based on the results, comparisons are made under pooled/unpooled variance

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

for the difference between the means of the two groups are shown in Table 4. Both beginner and intermediate level students generally agreed on the positive aspects of ChatGPT ($M < 3.0$). The results of the t-test comparing the differences between the student groups show that intermediate level students have a more positive perception of the five questions A06, A07, A08, A09, and A12 ($p < 0.05$).

Ten questions about the negative aspects of using ChatGPT were asked. The detailed questions are described in Appendix 1. Respondents answered each question on a 5-point scale ranging from strongly agree (1) to strongly disagree (5). The means of the responses by student group are shown in Fig. 14, and the t-test results for the difference between the means of the two groups are shown in Table 5. Compared to intermediate level students, beginner level students were generally more concerned about the possible negative effects of ChatGPT. The results of the t-test comparing the differences between the student groups show that beginner level students have a more negative perception of nine questions, except D10 ($p < 0.05$).

5 Conclusions

This study aimed to investigate whether ChatGPT is indeed an educational tool by conducting research with beginner and intermediate level students using the Python programming language. For beginners in programming, the assistance from ChatGPT might be limited due to their lack of foundational knowledge. In contrast, intermediate learners, with pre-existing programming skills, could better utilize ChatGPT to enhance their learning process. This aligns with the Prior Knowledge Activation theory, where the utility of ChatGPT seems to vary based on the learner's existing programming knowledge. The study underscores the importance of not overly depending on immediate responses from ChatGPT and emphasizes the need

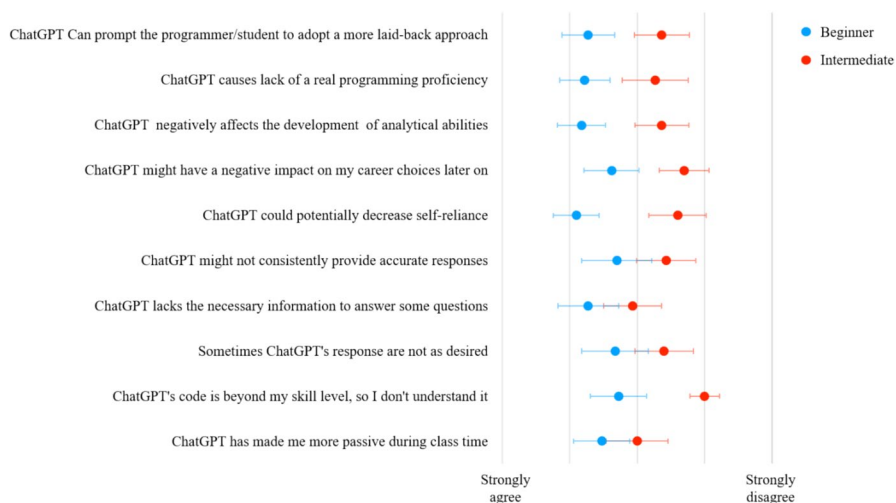


Fig. 14 Perceptions of disadvantages of ChatGPT by student group

Table 5 Results of student group comparison on negative perception of ChatGPT

Questions	Beginner Level		Intermediate Level		Variance Assumption	<i>p</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
D01	2.275	1.219	3.367	1.098	Equal	0.000***
D02	2.225	1.165	3.267	1.311	Equal	0.001**
D03	2.175	1.107	3.367	1.066	Equal	0.000***
D04	2.625	1.275	3.700	0.988	Equal	0.000***
D05	2.100	1.057	3.600	1.133	Equal	0.000***
D06	2.700	1.620	3.433	1.165	Equal	0.039*
D07	2.275	1.414	2.933	1.143	Equal	0.041*
D08	2.675	1.542	3.400	1.163	Equal	0.035*
D09	2.725	1.301	4.000	0.587	Unequal	0.000***
D10	2.475	1.301	3.000	1.232	Equal	0.092

The assumption of equal variances is confirmed using the F-test with a significance level of 0.05. Based on the results, comparisons are made under pooled/unpooled variance

* $p < 0.05$. ** $p < 0.01$. *** $p < 0.001$

for students to continue developing critical thinking, problem-solving skills, and self-directed learning abilities.

Furthermore, experts' evaluation also revealed differences. Beginners sometimes included content in their assignments that they hadn't learned, suggesting that ChatGPT did not contribute significantly to improving their programming skills. In contrast, the homework codes of intermediate users demonstrated high readability and originality, aspects that were reflected in their mid-term exam performances. These results show significant differences in the influence and utilization of ChatGPT between beginners and intermediate users. As a result, for intermediate learners, ChatGPT seems to be helpful in improving programming skills, but it does not appear to be as beneficial for beginners in enhancing their programming abilities.

Survey results showed distinct reasons for using ChatGPT among beginners and intermediates. Beginners found it easy and confidence-boosting, while intermediates used it for debugging and finding alternative solutions. However, beginners were concerned about potential negative impacts on their future careers, a worry less prevalent among intermediates.

5.1 Discussion

The integration of generative AI into education has transformed the learning landscape, offering tools that provide instant feedback, personalized assistance, and access to vast amounts of information. However, the experimental findings reveal that learners with insufficient prior knowledge may struggle to effectively leverage generative AI as a learning tool. This issue arises from their inability to critically evaluate, apply, or adapt the solutions provided by AI. Instead, they may passively

accept the AI's outputs, which can hinder deep learning and the development of problem-solving skills.

Proportionality refers to the balance between the resources provided and the learner's capacity to utilize them effectively. In the context of generative AI, the responses and solutions it offers must align proportionally with the learner's existing knowledge and cognitive abilities. When this proportionality is disrupted—such as when AI provides advanced solutions to novices without adequate scaffolding—learners may become overwhelmed, leading to a reduced capacity to achieve meaningful learning outcomes. This ties in with the survey findings that beginner students are more concerned about the potential negative effects of ChatGPT.

For instance, providing generative AI tools to beginners in programming without adequate foundational support is akin to giving advanced machinery to an untrained worker; the tools, while powerful, may not translate into effective usage. The principle of proportionality highlights the need for educators to tailor AI usage to the learner's skill level. By doing so, learners can incrementally build their knowledge, maintaining a manageable cognitive load and fostering productive engagement with AI-generated content.

5.2 Implications

ChatGPT has found extensive applications in various educational areas, including individualized tutoring, writing assistance, language learning, and concept explanation. However, its reliability isn't absolute, especially in problem-solving tasks. Educators often caution against an overdependence on immediate responses from AI tools like ChatGPT. This research contributes to this discourse by exploring the usage of ChatGPT among students at different learning stages in Python programming.

The research reinforces the view of ChatGPT as a supplemental tool in education rather than a standalone solution. It emphasizes the importance of students continuing to develop critical skills such as logical thinking, problem-solving, and self-directed learning. The effective use of technology in education, therefore, lies in striking a balance between leveraging AI tools for assistance and fostering independent learning and critical thinking.

To maximize the benefits of generative AI while mitigating its drawbacks, several educational strategies can be adopted. Before introducing generative AI, educators should assess students' prior knowledge and provide foundational instruction as needed. For programming education, this might include teaching basic syntax and logic before allowing students to use AI for programming assistance. And

developing students' AI literacy is critical as well. Learners should be taught how to critically evaluate AI-generated content, recognize potential inaccuracies, and use AI as a collaborative tool rather than a definitive source of truth. Educators should provide scaffolding to support learners' use of generative AI tools. This could involve integrating prompts that encourage students to critically analyze AI outputs, cross-reference with reliable sources, or explain the reasoning behind AI-generated solutions.

5.3 Limitations and future research

This research has some limitations, and it is important to consider these limitations in relation to future research. First, this study was confined to Python programming, which restricts the ability to generalize the findings to a broader educational context. Future studies should explore the impact of ChatGPT on other programming languages and even extend similar methodologies to different academic disciplines. This expansion could provide valuable insights into ChatGPT's utility across diverse learning environments.

Second, the study did not account for factors like cultural differences, variations in educational systems, and individual learner characteristics, which could significantly influence the effectiveness of ChatGPT. Future research should delve into these aspects to offer a more holistic understanding of ChatGPT's role in education. Third, research assumed that university students, categorized as either beginner or intermediate Python users, would readily use ChatGPT software without resistance. Future studies must validate this assumption and thoroughly assess control variables for more accurate findings.

Fourth, the survey was conducted with a small sample size of forty beginner and thirty intermediate users of ChatGPT. To ensure more accurate comparisons and a comprehensive analysis, future research should involve a larger and more diverse sample size. As a result, future research should expand in scope, methodology, and sample size to address the limitations and provide a broader understanding of ChatGPT's potential and limitations in various educational settings.

Last, this study did not address how students used ChatGPT; the prompts they gave to ChatGPT are also important enough that there is an entire field of research called prompt engineering for effective use of ChatGPT. For example, asking ChatGPT to "Solve this problem" versus "Would you teach me how to solve this problem" yields very different results. In the latter case, ChatGPT provides a detailed step-by-step solution, which can be very helpful for beginners. To use ChatGPT as an educational tool, our research can be extended to design instructions on how to use it well.

Appendix 1: Questionnaire

Appendix A: Questionnaire

Sociodemographic Information

Q1. What gender do you identify as?

Q2. What is your highest level of education?

Q3. In the past week, on average, how many times have you used ChatGPT?

Viewpoints on the Use of ChatGPT (Multiple answers allowed)

Q4. What aspect of ChatGPT do you find the most appealing?

Cost	Time Efficiency	Infinite Creativity	Consistency
Ease of Use	High-quality	Flexibility	Accuracy

Q5. What is the most compelling feature of ChatGPT in the context of programming?

Using Natural Language	Easy Access	Quick Response	Personalized Learning
Multi-Language Support	Clear Explanations	Examples and Applications	
Inquiry and Search	Advanced Topics	Debugging and Feedback	

Advantages of ChatGPT (5-point scale of disagree to agree)

A01. ChatGPT enhances programming abilities.

(1) Strongly Agree (2) Agree (3) Neutral (4) Disagree (5) Strongly Disagree

A02. ChatGPT nurtures critical thinking skills.

(Same as A01)

A03. ChatGPT boosts confidence in programming.

(Same as A01)

A04. ChatGPT demonstrates how to write code more accurately and appropriately.

(Same as A01)

A05. ChatGPT enables the solving of complex problems.

(Same as A01)

A06. ChatGPT helps to avoid time wastage during research.

(Same as A01)

A07. ChatGPT aids in debugging.

(Same as A01)

A08. ChatGPT presents alternative problem-solving approaches.

(Same as A01)

A09. ChatGPT responds quickly and efficiently to questions.

(Same as A01)

A10. ChatGPT provides guidance and ideas for project creation.

(Same as A01)

A11. ChatGPT offers solutions for interdisciplinary issues like math and physics that are integral to the programming process.

(Same as A01)

A12. ChatGPT helps when I can't continue while writing code.

(Same as A01)

Disadvantages of ChatGPT (5-point scale of disagree to agree)

D01. ChatGPT Can prompt the programmer/student to adopt a more laid-back approach.

- (1) Strongly Agree
- (2) Agree
- (3) Neutral
- (4) Disagree
- (5) Strongly Disagree

D02. ChatGPT causes lack of a real programming proficiency.

(Same as D01)

D03. ChatGPT negatively affects the development of analytical abilities.

(Same as D01)

D04. ChatGPT might have a negative impact on my career choices later on.

(Same as D01)

D05. ChatGPT could potentially decrease self-reliance.

(Same as D01)

D06. ChatGPT might not consistently provide accurate responses.

(Same as D01)

D07. ChatGPT lacks the necessary information to answer some questions.

(Same as D01)

D08. Sometimes ChatGPT's response are not as desired.

(Same as D01)

D09. ChatGPT's code is beyond my skill level, so I don't understand it.

(Same as D01)

D10. ChatGPT has made me more passive during class time.

(Same as D01)

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Author contribution **Jihae Suh**: Writing—original draft, Conceptualization, Data curation, Methodology. **Kyuhan Lee**: Conceptualization, Investigation. **Jaehwan Lee**: Formal analysis, Visualization, Writing—review & editing.

Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interest The authors have no relevant financial or non-financial interests to disclose.

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